

EX:1	SETTING UP PYTHON ENVIRONMENT AND LIBRARIES – JUPYTER NOTEBOOK
DATE:	

AIM:

To set up the Python programming environment by installing Python, Jupyter Notebook, and the required libraries, and to verify that the environment is ready for Python programming and data analysis.

ALGORITHM:

Step 1: Install Python or Anaconda on the computer.

Step 2: Open the Command Prompt or Terminal.

Step 3: Verify the Python installation.

Step 4: Install Jupyter Notebook.

Step 5: Install the required Python libraries such as NumPy, Pandas, Matplotlib, and Scikit-learn.

Step 6: Launch Jupyter Notebook.

Step 7: Create a new Python Notebook.

Step 8: Import the required libraries.

Step 9: Execute the notebook to verify that all libraries are installed successfully.

Step 10: Save the notebook and close Jupyter Notebook.

PROGRAM:

```
print("Welcome to Data Analytics Lab")
```

```
a = 20
b = 10

print("Addition =", a+b)
print("Subtraction =", a-b)
print("Multiplication =", a*b)
print("Division =", a/b)
```

```
name = "John"
dept = "AI&DS"
cgpa = 8.9

print("Name:", name)
print("Department:", dept)
print("CGPA:", cgpa)
```

```
import numpy as np

a = np.array([10,20,30,40,50])

print(a)
print("Mean =", np.mean(a))
print("Sum =", np.sum(a))
```

```
import pandas as pd

data = {
    'Name': ['John', 'David', 'Alex'],
    'Marks': [85, 90, 95]
}

df = pd.DataFrame(data)

print(df)
```

```
import matplotlib.pyplot as plt

x = [1,2,3,4,5]
y = [10,20,15,25,30]

plt.plot(x,y)
plt.title("Sample Line Graph")
plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.show()
```

OUTPUT:

Welcome to Data Analytics Lab

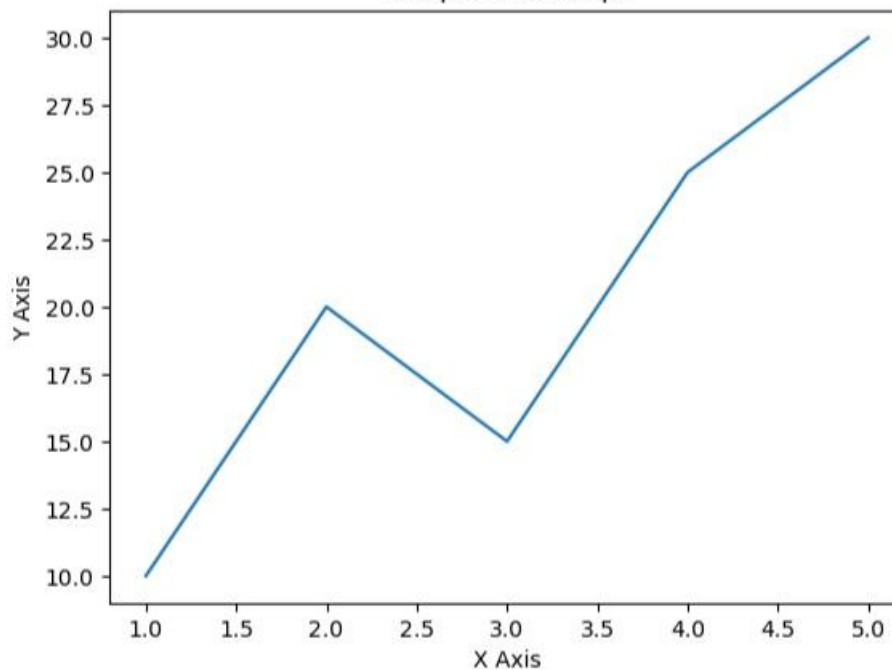
Addition = 30
Subtraction = 10
Multiplication = 200
Division = 2.0

Name: John
Department: AI&DS
CGPA: 8.9

[10 20 30 40 50]
Mean = 30.0
Sum = 150

	Name	Marks
0	John	85
1	David	90
2	Alex	95

Sample Line Graph



RESULT:

The Python environment, Jupyter Notebook, and the required libraries were successfully installed and verified.